

A Fraction of a Whole Number or Set

Answer Key

Grade 4–5 Multiplying and Dividing Fractions

Quick Answers

1. 6	4. 24	7. 0.3	10. 27
2. 7	5. 40	8. 16	
3. 12	6. 35	9. 256	

Curriculum

Level: Grade 4–5 Multiplying and Dividing Fractions

5.NF.B – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 10 tagged checks.

Common wrong answers

- 3. *If they got 27: divided by the top number and multiplied by the bottom one*
 - 4. *If they got 8: stopped after making the groups and reported just one group*
 - 7. *If they got 19.2: multiplied by 8 instead of splitting the 2.4 litres into 8 equal parts*
 - 8. *If they got 12: answered how many DO ride the bus instead of how many do not*
 - 10. *If they got 9: made the five equal groups but then took only one of them*
-

1. $\frac{1}{4}$ of 24 crayons.

Step 1. The bottom number is 4, so break the 24 crayons into 4 equal groups. On the picture, three rows of eight ring neatly as four groups of six — any grouping works as long as the groups match.

Step 2. $24 \div 4 = 6$, so one group holds 6 crayons.

Step 3. The top number is 1, so Ravi takes one of those groups.

6 crayons

Teaching note: say the sentence back — “one fourth of twenty-four is six” — and check it the other way: four groups of six rebuild 24.

2. $\frac{1}{3}$ of 21 stickers.

Step 1. The bottom number is 3: break the 21 stickers into 3 equal groups. The picture is three rows of seven, so each row is already one group.

Step 2. $21 \div 3 = 7$, so one group holds 7 stickers.

Step 3. One group goes to her brother.

7 stickers

Listen for: counting the rows instead of the stickers in a row. The answer is how many are *in* one group, not how many groups there are.

3. $\frac{2}{3}$ of 18 marbles.

Step 1. Divide first. The bottom number 3 splits the bar into 3 equal parts, and $18 \div 3 = 6$, so each part is worth 6 marbles.

Step 2. Multiply second. The top number 2 says to shade 2 of those parts: 2×6 .

12 marbles

Step 3. Check on the picture: two shaded parts of 6 and one unshaded part of 6 rebuild the 18 ✓

Common wrong answer: 27 — divided by the top number and multiplied by the bottom one ($18 \div 2 \times 3$). The bottom number always makes the groups.

4. $\frac{3}{4}$ of 32 pages.

Step 1. Divide: $32 \div 4 = 8$, so each of the 4 equal parts of the book is 8 pages.

Step 2. Multiply: Dev has read 3 of those parts, 3×8 .

24 pages

Step 3. Check: the one part he has not read is 8 pages, and $24 + 8 = 32$ ✓

Common wrong answer: 8 — the groups were made correctly and then only one was counted. The top number is the instruction to take three.

5. Working backwards: one fifth of the bag is 8 grapes.

Step 1. This time the *part* is known and the whole is missing. The bottom number 5 says the bag is 5 equal parts, and one of them is 8 grapes.

Step 2. Write 8 in every part of the bar — all five parts are the same size, which is what “equal parts” means.

Step 3. The whole bag is all five parts: 5×8 .

40 grapes

Step 4. Check forwards: one fifth of 40 is $40 \div 5 = 8$ ✓

Teaching note: forwards you divide, backwards you multiply. Drawing the bar first is what keeps the two directions straight.

6. $\frac{5}{6}$ of a 42-minute period.

Step 1. Divide: $42 \div 6 = 7$, so each of the 6 equal parts of the period is 7 minutes.

Step 2. Multiply: the project takes 5 of those parts, 5×7 .

35 minutes

Step 3. Check: the leftover part is 7 minutes of tidying, and $35 + 7 = 42$ ✓

Teaching note: nothing changes when the “set” is time instead of objects — 42 minutes still breaks into six equal sevens.

7. $\frac{1}{8}$ of a 2.4 litre jug.

Step 1. The bottom number 8 still means equal groups, so split the 2.4 litres into 8 equal parts.

Step 2. $2.4 \div 8 = 0.3$, and the top number is 1, so one part is the answer.

0.3 L

Step 3. Check: eight glasses of 0.3 litres is $8 \times 0.3 = 2.4$ litres ✓

Common wrong answer: 19.2 — the jug was multiplied by 8 instead of split into 8 parts. A quick sense check kills it: one glass cannot hold more than the whole jug.

8. 28 students, $\frac{3}{7}$ ride the bus — how many do not?

Step 1. Find the part the fraction names first. Divide: $28 \div 7 = 4$, so each of the 7 equal groups is 4 students.

Step 2. Multiply: $3 \times 4 = 12$ students ride the bus.

Step 3. The question asks for the rest, so subtract from the whole class: $28 - 12$.

16 students

Step 4. Check: $12 + 16 = 28$ ✓ (The students who do not ride are $\frac{4}{7}$ of the class, and $4 \times 4 = 16$ as well.)

Common wrong answer: 12 — the fraction was computed correctly and the last sentence of the question was not read. Worth underlining the word *not* together.

9. Working backwards: $\frac{3}{8}$ of the theatre is 96 people.

Step 1. The shaded part of the bar is 3 eighths, and those 3 parts together hold 96 people.

Step 2. Find one part: $96 \div 3 = 32$ seats in one eighth.

Step 3. The whole theatre is 8 of those parts: 8×32 .

256 seats

Step 4. Check forwards: $256 \div 8 = 32$ and $3 \times 32 = 96$ ✓

Common wrong answer: dividing 96 by 8 to get 12. The 96 covers three parts, not one, so it has to be shared between three before the whole is rebuilt.

10. What Sam got right and what he forgot.

Step 1. What he did right: the bottom number is 5, so splitting the 45 counters into five equal groups is exactly the first move, and $45 \div 5 = 9$ counters per group is correct.

Step 2. What he forgot: the top number. $\frac{3}{5}$ asks for *three* of those groups, and Sam stopped after pointing at one. His 9 is the answer to $\frac{1}{5}$ of 45, not $\frac{3}{5}$.

Step 3. Finish the job: 3×9 .

27 counters

Step 4. Check on the picture: three groups of 9 are shaded and two groups of 9 are not, and $27 + 18 = 45$ ✓

Full credit on the sentence: it has to name both halves — the equal groups were made correctly, and the top number was never used. “He got it wrong” does not show that the student can see where the method broke.